

Lab - 9

Apply Decision Tree for the given dataset.

Instance	a_1	a_2	a_3	Classification
1	True	Hot	High	No
2	True	Hot	High	No
3	False	Hot	High	Yes
4	False	Cool	Normal	Yes
5	False	Cool	Normal	Yes
6	True	Cool	High	No
7	True	Hot	High	No
8	True	Hot	Normal	Yes
9	False	Cool	Normal	Yes
10	False	Cool	High	Yes

Form entropy of all data

Distinct values in classification	Total
Yes	6
No	4

$$\begin{aligned}
 \text{Entropy}(D) &= \frac{-6}{10} \log_2\left(\frac{6}{10}\right) - \frac{4}{10} \log_2\left(\frac{4}{10}\right) \\
 &= 0.9709
 \end{aligned}$$

Gain of a_1 :-

$$\text{Gain}(D, a_1) = \text{Entropy}(D) - \text{Entropy}(a_1)$$

Distinct values in a_1	Yes	No	Total
True	1	4	5
False	5	0	5

$$\begin{aligned} \text{Entropy}(a_1) &= \frac{5}{10} \times \left[-\frac{1}{5} \log_2\left(\frac{1}{5}\right) - \frac{4}{5} \log_2\left(\frac{4}{5}\right) \right] + \\ &\quad \frac{5}{10} \times \left[-\frac{5}{5} \log_2\left(\frac{5}{5}\right) - \frac{0}{5} \log_2\left(\frac{0}{5}\right) \right] \\ &= 0.7219 + \frac{5}{10} \Rightarrow 0.3609 \end{aligned}$$

$$\begin{aligned} \text{Gain}(D, a_1) &= 0.9709 - 0.3609 \\ &= 0.6099 \end{aligned}$$

Gain of a_2 :-

Distinct values in a_2	Yes	No	Total
Hot	2	3	5
Cool	4	1	5

$$\begin{aligned} \text{Gain}(D, a_2) &= 0.9709 - \left\{ \frac{5}{10} \left[-\frac{2}{5} \log_2\left(\frac{2}{5}\right) - \frac{3}{5} \log_2\left(\frac{3}{5}\right) \right] \right. \\ &\quad \left. + \right\} \end{aligned}$$

$$\frac{5}{10} \left[\frac{-4}{5} \log_2 \left(\frac{4}{5} \right) - \frac{1}{5} \log_2 \left(\frac{1}{5} \right) \right] \}$$

$$= 0.1245$$

Gain for a_2 :

Distinct values of a_2	Yes	No	Total
High	2	4	6
Normal	4	0	4

$$\text{Gain}(D, a_2) = 0.9709 - \left\{ \frac{6}{10} \left[\frac{-2}{6} \log_2 \left(\frac{2}{6} \right) - \frac{4}{6} \log_2 \left(\frac{4}{6} \right) \right] + \frac{4}{10} \left[\frac{-4}{4} \log_2 \left(\frac{4}{4} \right) \right] \right\}$$

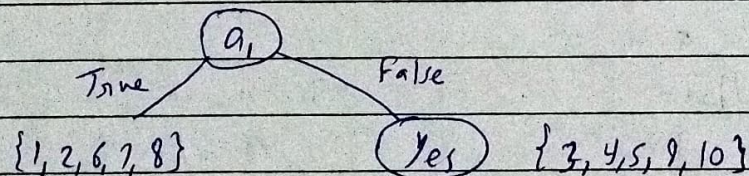
$$= 0.4200$$

$$\text{Gain}(D, a_1) = 0.6099 \rightarrow \text{Maximum}$$

$$\text{Gain}(D, a_2) = 0.1245$$

$$\text{Gain}(D, a_3) = 0.4200$$

$\Rightarrow$



New Data

Instance	a_2	a_1	Classification
1	Hot	High	No
2	Hot	High	No
6	Cool	High	No
7	Hot	High	No
8	Hot	Normal	Yes

$$\text{Entropy}(D) = -\frac{1}{5} \log_2\left(\frac{1}{5}\right) - \frac{4}{5} \log_2\left(\frac{4}{5}\right)$$

$$= 0.7219$$

Distinct values in Classification	Total
Yes	1
No	4

Gain of a_2

$$\text{Gain}(D, a_2) = \text{Entropy}(D) - \text{Entropy}(a_2)$$

Distinct values in a_2	Yes	No	Total
Hot	1	3	4
Cool	1	0	1

$$\begin{aligned} \text{Gain}(D, a_2) &= 0.7219 - \left\{ \frac{-4}{5} \left[\frac{1}{4} \log_2\left(\frac{1}{4}\right) - \frac{3}{4} \log_2\left(\frac{3}{4}\right) \right] \right. \\ &\quad \left. + \frac{1}{5} \left[\frac{-1}{1} \log_2\left(\frac{1}{1}\right) \right] \right\} \\ &= 0.0729 \end{aligned}$$

Gain for a_3

Distinct values of a_3	Yes	No	Total
High	0	4	4
Normal	1	1	1

$$\begin{aligned} \text{Gain}(D, a_3) &= 0.7219 - \left\{ \frac{4}{5} \left[\frac{-4}{4} \log_2\left(\frac{4}{4}\right) \right] + \right. \\ &\quad \left. \frac{1}{5} \left[\frac{-1}{1} \log_2\left(\frac{1}{1}\right) \right] \right\} \\ &= 0.7219 \end{aligned}$$

